

Scaling Full-Ternary-Secret NTRU Signatures: FT1536 as a Baseline for FT768 and FT3072

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Abstract

We study how security geometry, numerical stability and implementation cost scale across the family of full-ternary-secret NTRU signature profiles $N \in \{768, 1536, 3072\}$ over $R_N = \mathbb{Z}[X]/(X^N - X^{N/2} + 1)$ with $q = 18433$, taking the implemented FT1536 profile as baseline. Three contributions replace a withdrawn draft whose security tables were not estimator outputs. First, we give the exact metric transport that the verification predicate uses: the quadratic form computed by `falcon_is_short` is the normalized trace $Q_{A_2}(a) = \frac{1}{N} \text{Tr}_{K/\mathbb{Q}}(a\bar{a}) = \sum_{i < N/2} (a_i^2 + a_i a_{i+N/2} + a_{i+N/2}^2)$ with sharp equivalence constants $1/2 \leq Q_{A_2}/\|\cdot\|_2^2 \leq 3/2$, and we transport determinant and Gaussian-heuristic radii into this metric. Second, we prove and kernel-check the exact layout recurrences of the expansion tree, $T = (\log n + 2)N$ words, $\text{sk} = (\log n + 6)N$, leaf count N , giving tree sizes 8448/18432/39936 words for the three degrees. Third, we *propose* an input-norm-carrying forward-error bound for the source-order FFT3 (status: proposed, not proved) and check it is not refuted on twelve finite cases against the pinned twiddle tables. All attack-cost cells are `NOT_RUN`: the three problems (key recovery, MT-ISIS byte production, sampling law) are defined precisely and a pinned estimator campaign is provided but not executed. The honest conclusion is that the degree alone leaves geometry and layout parametrically controlled, while every numeric domain certificate and all attack costs remain per-instance obligations; FT768 and FT3072 are planned profiles, not implemented schemes.

1 Motivation and related work

FT1536 is a research signature profile with *full ternary secrets*: the NTRU key polynomials f, g have all coefficients in $\{-1, 0, 1\}$, unlike the Gaussian-like sampling of the classical Falcon design [7]. The project treats FT1536 as a heavyweight profile for infrequent operations on important objects (data blocks, manifests, checkpoints); FT768 is a lighter compromise and FT3072 a long-term research direction. None of the planned profiles is a finished scheme; the pinned sources reject every ternary degree but $N = 1536$ (Section 2).

This manuscript is the corrected second iteration of a scaling study. The first submission was not accepted with cause: its security tables were produced by an ad-hoc formula, not by a lattice estimator; it conflated the Gaussian-like key population of a historical worksheet with the current ternary family; and it claimed floating-point sufficiency without proof. All such numbers are withdrawn together with their dependent conclusions; the present text reports only what is established, and states `NOT_RUN` or `OPEN` everywhere else.

Related work used here: the Falcon specification and its sampling and key generation framework [7, 11, 8]; NTRU subfield and overstretched-parameter attacks and their refinements [1, 10, 6]; hybrid attacks [9, 14]; the core-SVP cost model of Alkim–Ducas–Pöppelmann–Schwabe [4] as implemented in the lattice estimator [2, 3]; Rényi-divergence accounting for sampler loss [13, 12]; and recent fixed-point implementation work for Falcon [5], which is the closest external analogue of our integer-emulated arithmetic (FPEMU). Positioning: the FT family of `free_falcon_sign` is an independent research family; no standards-body categories

are claimed, mapped or cited as normative for it (owner decision, 2026-09-22). Falcon-512/1024 appear below strictly as research reference points measured with the same ruler.

2 Family definition and scaling regimes

2.1 The family

For $N = 3 \cdot 2^k$ let

$$R_N = \mathbb{Z}[X]/(\Phi_N), \quad \Phi_N(X) = X^N - X^{N/2} + 1, \quad q = 18433.$$

Lemma 1 (cyclotomic identity). *For $N = 3 \cdot 2^k$ one has $\Phi_N = \Phi_{3N}$, the $(3N)$ -th cyclotomic polynomial; in particular Φ_N is irreducible of degree $N = \varphi(3N)$.*

Proof sketch; exact coefficient check in the artifacts. $\Phi_6(Y) = Y^2 - Y + 1$, so $\Phi_N = \Phi_6(X^{N/2})$. If ζ is a primitive m -th root with $m = 3N$, then $\zeta^{N/2}$ has order 6 exactly when $\gcd(N/2\text{-shift}, m)$ forces $m = 3N$; more elementarily $\zeta^N - \zeta^{N/2} + 1 = \omega - w + 1 = 0$ with $\omega = e^{2\pi i/3}$, $w = e^{\pi i/3}$, and a degree count closes the claim. Coefficients of $\Phi_{2304}, \Phi_{4608}, \Phi_{9216}$ were compared over $\mathbb{Z}$ (sage_exact.json, SA1: all equal). $\square$

Since $3N \mid q - 1 = 2^{11} \cdot 3^2$, the ring has explicit evaluation (NTT) bases of $3N$ -th roots. Correct primitive roots for the three degrees are 3532, 625, 25 respectively (kernel-checked: $\Phi_N(w) = 0 \bmod q$ and order exactly $3N$). The values 8, 27, 13 appearing in the withdrawn draft have order N and give $\Phi_N(w) = 3 \bmod q$: for $\text{ord}(w) = N$ one gets $w^{N/2} = -1$, hence $\Phi_N(w) = 3$. They are usable only with an explicit twist. Our recomputations use the slot-to-root map $r_j = 1 + 6 \text{rev}_{k-1}(j/3) + N(j \bmod 3)$ of the source transform ($\text{rev}_{\log n - 2} = \text{rev}_{k-1}$ is the bit reversal over $\log n - 2$ bits; one bit fewer would give only 384 distinct roots at $N = 1536$ instead of 768).

What is implemented. The pinned candidate (17-file manifest 56974571...) contains the FT1536 profile only: `load_skey` and the key certificate reject $\log n \neq 10$, and the tree size is hard-coded as $12n$ (valid only at $\log n = 10$). Test vectors for a historical degree-768 ternary variant exist; per the assignment they are *not* the designed FT768. Table 1 lists the baseline parameters and the two planned degrees.

Table 1: Baseline (pinned) and planned profiles. B is the strict verification threshold in the normalized-trace metric; σ is the candidate signing width. Only the FT1536 row is pinned by artifacts.

	FT768	FT1536 (baseline)	FT3072
$N = 3 \cdot 2^{\log n - 1}$	768	1536	3072
conductor $3N$	2304	4608	9216
q	18433	18433	18433
secrets f, g	raw proposal iid uniform in $\{-1, 0, 1\}$; emitted keys conditioned on KeyGen		
σ	OPEN	768 (SIG-001 OPEN)	OPEN
B	OPEN	2093922385	OPEN
attempt caps		KeyGen $3 \cdot 10^6$, Sign 16; $2N$ scalar samples per attempt	
status	planned	implemented	planned

The baseline threshold is exactly $B = \lfloor (43/40)^2 \cdot 2N \cdot 768^2 \rfloor = 2093922385$ (kernel-checked); $2N \cdot 768^2 = 1811939328$ is the margin-less expectation that the withdrawn table had used as a threshold. The source itself marks the choice $\sigma = 768$, margin 1.075 as open (comment SIG-001).

2.2 Two scaling regimes

Regime A (change the degree only). Keep q , the ternary proposal law and the current arithmetic; enumerate what ceases to work: the $\log n$ -specific hard checks reject other degrees; the adaptive CDF bank is fixed (5 proposal levels $\sigma_0^2 \in \{5, 20, 80, 320, 768\}$, 512 entries, proposal support $[-365, 366]$ certified for the FT1536 instance only); the per- $\log n$ candidate thresholds of `falcon_is_short` have no derivation in the pinned sources (their ratio bound/ $(2N)$ moves in an arithmetic-like sequence of step ≈ 1608.2 , consistent with a historical May-2026 value 160982450, and the baseline breaks the sequence); all H3 numeric domains are per-instance certificates. Under Regime A, FT768/FT3072 are *not* defined schemes.

Regime B (co-designed parameters). Admit q_N, σ_N, B_N together with precision and table choices. Parameter selection must follow a stated measurable criterion; we use, in this order: (i) *acceptance mass*: per-attempt rejection at the level certified in the ideal normalized-canonical model ($Q/\sigma^2 \sim \chi_{2N}^2$, i.i.d. Gaussian before casts/retries). With $x = B/(2\sigma^2) = 2093922385/1179648$ the tail is exactly $\Pr[Q \geq B] = e^{-x} \sum_{k=0}^{1535} x^k/k!$ (the Erlang/Poisson identity $\Pr[T \geq v] = e^{-v/2} \sum_{j < m} (v/2)^j/j!$ for $T \sim \text{Gamma}(m, 1/2)$, $m = 1536$), recomputed rigorously with exact $\mathbb{Z}/\mathbb{Q}$ sums and a `RealBallField` e^{-x} enclosure as $[2.9925 \dots 10^{-9} \pm \sim 10^{-305}]$, matching the independently certified interval $[2.9925 \dots 10^{-9} \pm 2.3 \cdot 10^{-82}]$ digit for digit. The resulting $2^{-28.32 \dots}$ refutes the earlier 2^{-40} demand *in that model*. The 2^{-28} figure is a PROPOSED model-level per-attempt target only — not an approved project goal and not a security level; the actual post-cast law remains OPEN; (ii) *geometry*: the key-vector ratio of Section 3 stays within a fixed factor across the family; (iii) *attack budget*: a pinned estimator campaign defines the target. Only after (i)–(iii) may σ_N, B_N be read off. As conditional algebra, if one imposes $\sigma_N \propto \sqrt{N}$ and $B_N = \lfloor 1.075^2 \cdot 2N\sigma_N^2 \rfloor$ then $B_N \propto N^2$; this identity alone proves nothing about KeyGen feasibility, smoothing, acceptance probability or signing-law accuracy. All values other than the baseline are labelled `ALTERNATIVE_PROPOSAL`.

2.3 Parameter matrix

parameter	constant	scaled	needs new proof
q	baseline / A	q_N (B)	NTT roots, gates
ring Φ_{3N}	by degree	—	Lemma 1 (done)
f, g law	ternary iid	—	conditioning facts (open)
σ, B	—	σ_N, B_N	SIG-001; acceptance mass
sampler tables	A: fixed	B: retuned	sampler-law loss per N
FPEMU precision	A: fixed	B: retuned	full error recurrences
tree layout	—	$(\log n + 2)N$	done (Thm. 1)

A comparable family *does* emerge from these choices only if Regime B criteria (i)–(iii) are all closed; today the matrix shows exactly which obligations are open.

3 Geometry and attack estimates

3.1 Metric transport

Let $K = \mathbb{Q}(\zeta_{3N})$ and embed R_N by the N complex embeddings, grouped into $N/2$ conjugate pairs σ_j . Write Q_{A_2} for the quadratic form actually computed by verification.

Lemma 2 (trace form of the verification metric). *For $a \in R_N$,*

$$\begin{aligned} Q_{A_2}(a) &= \frac{1}{N} \sum_{i,j < N} a_i a_j c_{3N}(i-j) = \frac{1}{N} \text{Tr}_{K/\mathbb{Q}}(a\bar{a}) = \frac{2}{N} \sum_{j=1}^{N/2} |\sigma_j(a)|^2 \\ &= \sum_{i < N/2} (a_i^2 + a_i a_{i+N/2} + a_{i+N/2}^2), \end{aligned}$$

where c_m are Ramanujan sums. The last expression is literally the loop of `falcon_is_short` (pair offset $N/2$).

Proof. $\text{Tr}(\zeta^t) = c_{3N}(t)$ and $c_m(t) = 0$ unless $m/(m, t)$ is squarefree; for $m = 9 \cdot 2^j$ and $0 < |t| < N$ this forces $N/2 \mid t$, with $c_m(0) = N$, $c_m(\pm N/2) = N/2$ (verified exactly for all three m). The quadratic form therefore collapses to $N/2$ blocks $x^2 + xy + y^2$ over pairs $(a_i, a_{i+N/2})$, and $x^2 + xy + y^2 = \frac{1}{2}(x^2 + y^2 + (x+y)^2)$ closes the middle identities. $\square$

Lemma 3 (equivalence constants and single embedding). *For all a , $\frac{1}{2}\|a\|_2^2 \leq Q_{A_2}(a) \leq \frac{3}{2}\|a\|_2^2$ with both extremes attained (a all-ones attains $3/2$; $a_i = -a_{i+N/2}$ attains $1/2$), and $\max_j |\sigma_j(a)|^2 \leq \frac{N}{2} Q_{A_2}(a)$.*

Proof. Each block gives $x^2 + y^2 \leq 2(x^2 + xy + y^2) \leq 3(x^2 + y^2)$, since the differences are $(x+y)^2$ and $(x-y)^2$ (kernel-checked). The embedding claim is Lemma 2 plus one term of the sum. $\square$

The “single embedding” heuristic used in the withdrawn draft is therefore *false* without the factor $N/2$: for $a = 1 + X + \dots + X^{N-1}$ and a primitive $3N$ -th root, $|a(\zeta)| = \sqrt{3}/(2 \sin(\pi/(3N)))$, which at $N = 1536$ equals $1270.261873\dots$ against the withdrawn bound $\sqrt{N} \max |a_i| = 39.19$ — a factor 32.4 violation (22.9 and 45.8 at $N = 768, 3072$).

3.2 Determinants, Gaussian heuristic, key statistics

The NTRU lattice $\Lambda_h = \{(u, v) : u + hv = 0 \bmod (q, \Phi_N)\}$ has dimension $d = 2N$, coefficient-metric determinant q^N , and, by Lemma 2, Gram determinant $\det G = (3/4)^N$ for the Q_{A_2} metric on (z_1, z_2) , hence $\text{covol}_{Q_{A_2}}(\Lambda_h) = (3/4)^{N/2} q^N$, i.e. $\text{covol}^{1/d} = (3/4)^{1/4} \sqrt{q}$. We use the exact radius convention $\text{GH} = \Gamma(d/2 + 1)^{1/d} \det^{1/d} / \sqrt{\pi}$ (the familiar $\sqrt{d/2\pi e} \det^{1/d}$ is its asymptotic form; both are tabulated in the artifacts). For raw iid ternary proposals $\mathbb{E}\|f\|_2^2 = 2N/3$ and the key vector is $(g, -f)$ with $\mathbb{E}\|(g, -f)\|^2 = 4N/3$ in both metrics in expectation; per-sample the two metrics differ within Lemma 3. Table 2 collects the exact numbers.

Table 2: Geometry (exact GH convention; coefficient metric and Q_{A_2} metric). Key ratio = $\sqrt{4N/3}/\text{GH}$ for raw ternary proposals (emitted keys are a conditioned population and are not covered).

N	768	1536	3072
GH (coeff.)	1291.09	1823.56	2577.12
GH (Q_{A_2})	1201.49	1697.01	2398.28
key ratio (coeff.)	0.024785	0.024817	0.024834
key ratio (Q_{A_2})	0.026634	0.026667	0.026686
GH($2N$)/GH(N)	1.4124		1.4132

The key ratio converges to $\sqrt{4\pi e/(3q)} = 0.0248538421\dots$ (and the single- f variant to $\sqrt{2\pi e/(3q)} = 0.0175743203\dots$): at fixed q the ternary key stays roughly $40\times$ shorter than the Gaussian-heuristic radius for every degree, while $\text{GH} \propto \sqrt{N}$ ($\log \text{GH} = \frac{1}{2} \log N + O(1)$). This is a statement about geometry, not about attack cost.

3.3 Three problems, kept apart

(1) **Key recovery** (trapdoor recovery): given h , find any short $(u, v) \in \Lambda_h$, in particular $(g, -f)$; the secret law is *ternary* — never the signing σ . (2) **MT-ISIS / accepted bytes**: given h, c , produce any (z_1, z_2) in the coset with $Q(z_1, z_2) < B_N$, where Q is exactly Lemma 2. Since B_N exceeds the typical coset minimum by a wide margin (Table 2), this search has exponentially many solutions and its cost should be reported as finding *any* coset vector below the threshold (e.g. via exact CVP in the coset), not as a generic SIS number. (3) **Sampling law**: the distribution of honest outputs and its divergence accounting — not an attack-cost cell at all. The one-sided statement “Verify accepts $\Rightarrow$ short witness” is proved for the pinned candidate (L_V checkpoint); the converse and the byte-level law are separate obligations.

Attack estimates: NOT_RUN.

attack class	FT768	FT1536	FT3072
primal uSVP/BDD (core-SVP $2^{0.265\beta}$ q.)	NOT_RUN	NOT_RUN	NOT_RUN
dual / hybrid / MITM	NOT_RUN	NOT_RUN	NOT_RUN
subfield norm-down (7 index-2, 7 quadratic)	NOT_RUN	NOT_RUN	NOT_RUN
overstretched refinements	NOT_RUN	NOT_RUN	NOT_RUN

No estimator was run for this manuscript. A pinned campaign skeleton with the exact problem parameterizations (ternary $\mathcal{U}\{-1, 1\}$ secrets, threshold- B_N coset search) and its preconditions is part of the artifacts; per the review process, cells may change from NOT_RUN only to recorded campaign output. We note without numbers that at $q = 18433$ and $N \leq 3072$ the “overstretched” regime of $[1, 10, 6]$ is not expected to begin; verifying this per subfield is task (4) of the research notes. All seven index-2 subgroups and all seven quadratic subfields were enumerated exactly (group census: 142/164/186 subgroups in total).

4 Numerical stability and sampling

The arithmetic backend is FPEMU: integer-emulated IEEE-754 binary64 with one rounding per primitive operation. The central question is whether the correlation-preserving normalization technique (raw- L /stable- D) gives *parametric* control at other degrees. The honest answer splits: the *structure* is parametric, the *certified domains* are not. Every numeric envelope of the H3 chain ($\text{div} \in [1/16, 2^{35}]$, $|\Im D_C| < 32$ at root and < 1 one level down, raw/stable factor < 6 , the weighted bank bound 849346588, margins 1209616765/1209616764) is an $N = 1536$ instance certificate; at $N = 768$ or 3072 each is a new obligation. We propose one parametric recurrence with its input norm restored (the withdrawn table computed $\varepsilon\sqrt{N}\log_2 N$ while its own lemma carried $\|x\|$). Its status is a *proposed bound* — plausible direction of argument, not a completed analytic result.

Proposition 1 (FFT3 forward error — proposed bound, to be proved). *Let $\text{fl}(a)$ be the source-order $\text{falcon_FFT3}(a, \log n, 1)$ in binary64 with twiddle constants of relative accuracy ε_{tw} . Then every output slot j satisfies*

$$|\text{fl}(a)_j - a(\zeta_{r_j})| \leq \gamma_{cl} (1 + \varepsilon_{tw})^{\ell+1} \|a\|_1, \quad \gamma_t = \frac{tu}{1 - tu}, \quad u = 2^{-53}, \quad \ell = \log n,$$

with c the per-level path length of the recursion (1 first pass, $\log_2(t/3)$ doubling levels, one tripling level).

Argument sketch (not a derivation); status: proposed. Each output is a sum of input coefficients multiplied by unit-circle twiddle products along one recursion path; standard first-order

error accumulation gives one γ -unit per rounded operation on the path, and twiddle error contributes the $(1 + \varepsilon_{tw})^{\ell+1}$ factor. The dependence on ℓ is linear — this is the requested N -dependent parameter. A full step-by-step derivation bound to the FPEMU operation contracts is OPEN (see Section 6); in particular the concrete constant c , the full operation domains and the twiddle-error budget ε_{tw} are not pinned by this argument. *Phase/twiddle perturbation requires a separate derivation: the multiplier $(1 + \varepsilon_{tw})^{\ell+1}$ is an assumed placeholder, not a derived term* — replacing the exact twiddle products by perturbed ones changes the function being evaluated at each node of the recursion, so a first-order operation-count argument does not by itself carry over to ideal phases. $\square$

Validation (not a proof). A source-faithful binary64 port using the *pinned twiddle bit patterns* of `fpr-emulated.h` and the slot map r_j was checked on four input classes (ternary, Gaussian, all-ones, alternating) at all three degrees against a 256-bit MPFR oracle (`sage.all.RealField(256)`, correctly rounded cosine/sum, exact integer inputs; the port is *not* proven bit-identical to the FPEMU macros): $\max_j |\Delta_j| \leq 7.8 \cdot 10^{-13}$, and the observed constant $\max |\Delta|/(\ell u \|a\|_1) \leq 0.209$ (exact maximum 0.20850...) against the crude bound $8\ell u \|a\|_1$ in all 12 runs. The all-ones class is exactly the adversarial family of Section 3. Twelve finite cases establish *neither* a uniform constant c *nor* the operation domains *nor* any FPEMU contract; they only record that the proposed bound is not refuted on them.

What remains open. (a) Binding Proposition 1 to the bit-exact FPEMU contracts (the model above rounds like the FPC macros but their operation order may differ in the last ulp) and quantifying ε_{tw} from the tables; (b) error recurrences for raw LDL, stabilization, center updates, reconstruction and rounding, including cancellation and subnormal behaviour at the actual domains; (c) the sampling law after casts, its retry behaviour and Rényi losses over shared histories [13]; (d) any argument that the pre-cast values are safe *before* the later $Q < B$ cast — explicitly not claimed. For FT3072 we state the sufficiency *conditions* under which current precision would survive (all recurrences within the certified domains after retuning) and the scope of the new obligation; we do not promise the precision is sufficient.

5 Implementation costs

The expansion tree of `ffLDL_fft3` is fully parametric.

Theorem 1 (layout recurrences). *With $S(r) = (r + 1)2^r$ (inner subtree at $\log n = r$), the depth-1 node at $\log n = k+1$ stores $D(k) = 3 \cdot 2^{k+1} + 3S(k)$ words, the top node at $\log n = k+2$ stores $T(k) = 3 \cdot 2^{k+1} + 2D(k) = (k+4) \cdot 3 \cdot 2^{k+1} = (\log n + 2)N$ words, and the whole expanded key $sk = 4N + T = (\log n + 6)N$ words with exactly N leaf words. The temporary buffer is $7N$ words.*

Proof sketch. Induction over the recursion of `ffLDL_inner_fft3_keygen` (node = 2^r L-words + two children, base $2 + 2$ words) gives S ; the 3×3 level contributes $3 \cdot 2^{k+1}$ L-words and three inner trees; the trisection top contributes one root L-block and two depth-1 trees. The closed forms are kernel-checked (`FTLayout.lean`), including all instance values. $\square$

Every FT1536 entry reproduces the frozen, independently replayed `RAW_ASSEMBLY` certificate exactly (basis 6144, tree 18432, leaves 1536, internal L 16896, tmp 10752, scratch 3584, high-water 8192). The scratch recurrence counts both consumers of the t_2 base — the child recursion and the node’s own d_1 store of 2^r words — as $W(r) = \max(2^r + W(r-1), 2^{r+1})$, giving 1792/3584/7168 words at the three degrees with no residual discrepancy. The code constant `treesize` = $12n$ equals $(\log n + 2)N$ only at $\log n = 10$ — at $11n$ and $13n$ for the planned degrees — which is itself evidence that the tree code is instance-locked. Verification cost is

Table 3: Cost and memory layout (8-byte words). Times are NOT_RUN; the withdrawn pseudo-cycle table is not replaced by modelled numbers.

	FT768	FT1536	FT3072
tree words	8448	18432	39936
expanded key = $(\log n + 6)N$	11520	24576	52224
expanded key (KiB)	90	192	408
tmp buffer $7N$ (KiB)	42	84	168
leaf words	768	1536	3072
scalar samples / attempt	1536	3072	6144
sign attempts (cap)	16 (expected count OPEN)		
KeyGen attempt cap	$3 \cdot 10^6$		
KeyGen / Sign / Verify time	NOT_RUN	NOT_RUN	NOT_RUN

dominated by hashing and one polynomial product ($O(N \log N)$ words/operations class); key generation cost is dominated by big-integer NTRU solving and an unbounded (capped) retry loop, so the three operations do *not* scale alike; without measurements we report no times. A single-signer/many-verifier deployment amortizes key generation and expands verification count; the security – size – memory – latency trade-off cannot be settled before the NOT_RUN cells are filled.

SIMD/AVX vectorization is future work: vectorizing FPEMU semantics-preservingly (one rounding per scalar op) versus changing the backend (different rounding points) are different experiments, and no speedups are claimed.

6 Evidence and limitations

Evidence classes are kept apart throughout, and the artifact ledger records each claim’s class and scope. Kernel-checked (Lean 4, clean logs, no `sorry/native_decide`): the A2 identities and equivalence constants (Lemma 3 algebra), the layout recurrences and all word counts (Theorem 1), the root/order tables and the threshold arithmetic $B = 2093922385$. Exact computation (Sage/GAP and binary64 controls): the cyclotomic identity, the Ramanujan support of Lemma 2, trace-form instances, the subgroup census, the counterexample values and the FFT3 validation. Analytic in this text: Lemmas 1–3, the determinant and GH transports, Proposition 1 (sketch level). NOT_RUN: every attack cost and every timing. OPEN: sampling law and real-PRNG bridge, pre-cast safety, precision sufficiency for FT3072, SIG-001 parameter selection.

Hash agreement of text with any manifest is not mathematical checking; a timing measurement with no signal is not constant-time evidence. Nothing here certifies FT768/FT3072, claims a security level, or evaluates deployment in critical infrastructure.

7 Conclusions

What the degree alone changes: the geometry ratios and the whole memory layout are parametric and now closed (Theorem 1, Table 2); the cyclotomic/NTT structure scales cleanly with Lemma 1; a *proposed* bound (Proposition 1, status: to be proved) captures the linear-in- $\log n$ error growth with a restored $\|a\|_1$ factor.

What needs joint parameter selection: σ_N , B_N and possibly q_N , under the stated acceptance/geometry/attack criteria; today only the baseline triple is pinned, and even there $\sigma = 768$ and the 1.075 margin are open (SIG-001).

What is actually established: the exact verification metric and its constant-equivalence to the trace form; exact root and subgroup tables; the layout theorem; one non-refuted proposed error bound (twelve finite cases, MPFR-checked); precise definitions of the three attack problems with their negative controls.

What must still be shown before calling FT768/FT3072 ready profiles: a pinned estimator campaign on the correctly defined problems (including the per-subfield analysis of all seven index-2 and seven quadratic subfields); new per-instance certificates for every H3 numeric domain at the new degrees; the sampling-law and real-PRNG obligations with Rényi accounting; retuned sampler tables and thresholds with derived, not fitted, constants; and measurements. A persuasive paper style does not substitute for any of these.

A Derivation notes

A.1 Ramanujan support. For $m = 9 \cdot 2^j$ and $0 < |t| < N$, the Ramanujan sum $c_m(t) = \mu(m/(m,t))\varphi(m)/\varphi(m/(m,t))$ vanishes unless $m/(m,t) \in \{1, 2, 3, 6\}$, i.e. $(m, t) \in \{m, m/2, m/3, m/6\}$, which for t in this range means $N/2 \mid t$; the surviving values are $c_m(0) = N$ and $c_m(\pm N/2) = N/2$. Expanding $\sum_{i,j} a_i a_j c_m(i-j)$ then yields Lemma 2.

A.2 Metric determinant. The Gram matrix of Q_{A_2} on (z_1, z_2) is $I_2 \otimes (I_{N/2} \otimes \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1 \end{bmatrix})$, of determinant $(3/4)^N$; volume scales by $\sqrt{\det} = (3/4)^{N/2}$ and $\det^{1/2N} = (3/4)^{1/4} \sqrt{q}$.

A.3 Bound arithmetic. $B = \lfloor (43/40)^2 \cdot 2N\sigma^2 \rfloor$ with $N = 1536$, $\sigma = 768$: numerator $1849 \cdot 1811939328$, denominator 1600, remainder 1472 — all kernel-checked.

B Artifact bindings

Source baseline: 17-file manifest `56974571b46e...` of Extra/c (FLOOR_CT candidate, identical across the examined commits). Recomputation entry points, relative to the package:

- kernel layer: `lean/FTA2.lean`, `lean/FTLayout.lean`, `lean/FTRoots.lean`, `lean/FTBounds.lean`;
- computation layer: `scripts/check_geometry.py`, `scripts/check_layout.py`, `scripts/check_bounds_table.py`, `scripts/check_fft3_error.py`, `scripts/sage_exact.sage`, with outputs in `results/` (`geometry.csv`, `layout.csv`, `sage_exact.json`, `fft3_error.json`, `bounds_table.json`);
- attack campaign: `scripts/estimator_campaign/` (status `NOT_RUN`).

The claim ledger is `CLAIMS.md`; the withdrawal ledger is `CORRECTIONS.md`.

References

- [1] Martin Albrecht, Shi Bai, and Léo Ducas. A subfield lattice attack on overstretched NTRU assumptions: Cryptanalysis of some FHE and graded encoding schemes. In *Advances in Cryptology – CRYPTO 2016*, 2016. Cryptology ePrint Archive, Paper 2016/127.
- [2] Martin Albrecht and contributors. Security estimates for lattice problems (the lattice estimator). Software repository, 2020–2026. Pinned commit `3e48ef421ec256afddb3e7d2249a77eab6e9ba12`, vendored and byte-verified against a fresh upstream clone.

- [3] Martin R. Albrecht, Rachel Player, and Sam Scott. On the concrete hardness of learning with errors. *Journal of Mathematical Cryptology*, 9(3):169–203, 2015. Cryptology ePrint Archive, Report 2015/046.
- [4] Edem Alkim, Léo Ducas, Thomas Pöppelmann, and Peter Schwabe. Post-quantum key exchange – A New Hope. In *25th USENIX Security Symposium (USENIX Security 16)*, pages 327–343. USENIX Association, 2016.
- [5] Daniel De Almeida Braga, Pierre-Alain Fouque, Bachir Lachguel, and Thomas Prest. Toward a secure fixed-point implementation of the Falcon signature scheme. In *CRYPTO 2026*, 2026. As listed on the authors’ publication page.
- [6] Léo Ducas and Wessel van Woerden. NTRU fatigue: How stretched is overstretched? In *Advances in Cryptology – ASIACRYPT 2021*, 2021.
- [7] Pierre-Alain Fouque, Jeffrey Hoffstein, Paul Kirchner, Vadim Lyubashevsky, Thomas Pornin, Thomas Prest, Thomas Ricosset, Gregor Seiler, William Whyte, and Zhenfei Zhang. FALCON: Fast-fourier lattice-based compact signatures over NTRU. Technical specification of the Falcon signature scheme (falcon-sign.info/falcon.pdf), 2017–2020. Author list as published at falcon-sign.info.
- [8] James Howe, Thomas Prest, Thomas Ricosset, and Mélissa Rossi. Isochronous Gaussian sampling: From inception to implementation. In *Post-Quantum Cryptography – PQCrypto 2020*, 2020.
- [9] Nick Howgrave-Graham. A hybrid lattice-reduction and meet-in-the-middle attack against NTRU. In *Advances in Cryptology – CRYPTO 2007*, 2007.
- [10] Paul Kirchner and Pierre-Alain Fouque. Revisiting lattice attacks on overstretched NTRU parameters. In *Advances in Cryptology – EUROCRYPT 2017*, 2017.
- [11] Thomas Pornin and Thomas Prest. More efficient algorithms for the NTRU key generation using the field norm. In *Public-Key Cryptography – PKC 2019*, 2019.
- [12] Thomas Prest. *Gaussian Sampling in Lattice-Based Cryptography*. PhD thesis, École Normale Supérieure, 2015.
- [13] Thomas Prest. Sharper bounds in lattice-based cryptography using the Rényi divergence. In *Advances in Cryptology – ASIACRYPT 2017*, 2017. Cryptology ePrint Archive, Report 2017/480.
- [14] Thomas Wunderer. A detailed analysis of the hybrid lattice-reduction and meet-in-the-middle attack. *Journal of Mathematical Cryptology*, 13(1):1–26, 2019.